Drone



Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Commercial drone; Different MODEL of drone; Hexacopter

What it is used for

A drone uses powered rotors, flight controls, and onboard electronics to perform approved aerial tasks. Pre-flight checks, controlled operation, and compliance with applicable aviation rules help reduce injury, equipment damage, and airspace risks.

Main parts

- Frame supports the drone's main components.
- Motors and propellers generate lift and movement.
- Flight controller stabilises the aircraft and manages control inputs.
- Battery supplies electrical power to the motors and electronics.
- Remote controller sends pilot commands to the drone.
- Navigation sensors and positioning systems support flight stability.
- Camera or mission payload performs the intended task when fitted.
- Landing gear supports the drone during take-off and landing.

Before you start

1. Confirm the operator is trained and authorised for the intended flight.
2. Check applicable local aviation rules, airspace restrictions, permissions, and weather conditions.
3. Inspect the frame, propellers, motors, landing gear, and fasteners for damage.
4. Check battery charge, secure mounting, and battery condition.
5. Verify controller operation, communication link, navigation status, and required software settings.
6. Choose a clear take-off and landing area away from people, vehicles, and obstacles.

How to use

1. Install the battery and payload according to the manufacturer's instructions.
2. Place the drone on a level surface and power it on in the specified sequence.
3. Wait for the required system checks and navigation status before flight.
4. Confirm the control response and set the approved flight mode and limits.
5. Take off gradually and hover at a safe, controlled height.
6. Operate the drone within visual and authorised operating limits as required by local rules.
7. Monitor battery level, signal quality, weather, and surrounding airspace throughout the flight.
8. Return to the designated landing area before the battery reaches its required reserve level.
9. Land slowly, disarm the motors, and power off in the recommended sequence.
10. Inspect the drone after flight, record any faults, and store the battery and equipment safely.

⚠ Safety

- Never fly near airports, restricted airspace, or emergency operations without required authorisation.
- Keep people clear of spinning propellers and never handle them while motors are armed.
- Do not fly in weather conditions beyond the drone's rated limits.
- Maintain required visual contact and follow applicable altitude and distance restrictions.
- Do not fly over people or sensitive sites unless specifically permitted under applicable rules.
- Stop the flight and land safely if control, navigation, battery, or communication problems arise.

Care & storage

- Clean the frame and sensors with suitable materials after use.
- Inspect propellers, motors, fasteners, and landing gear before and after flights.
- Store batteries according to the manufacturer's charge and temperature recommendations.
- Update firmware only through approved procedures and maintain flight and maintenance records.

Common mistakes to avoid

- Taking off without completing system and propeller checks.
- Flying with a damaged propeller or an insecure battery.
- Ignoring low-battery warnings, poor signal, or changing weather.
- Operating outside authorised airspace or losing required visual contact.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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